

ALY 6010 Prob Theory and Intro Stat – Report

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Introduction:

This report investigates the prediction of abalone age using readily accessible physical measurements, with the goal of simplifying the age determination process. The dataset used in this analysis consists of 4176 records, each containing measurements of various physical attributes of abalone, including shell dimensions, body weight, length, and ring count. The age of abalone is traditionally determined by counting the rings on their shells, a time-consuming and labor-intensive process. By establishing relationships between these physical attributes and abalone age, linear regression models were developed to facilitate more efficient age prediction.

Problem statement:

As age determination through microscopic examination of shell rings is time-consuming and expensive, developing accurate predictive models is crucial. Abalone, a rich and nutritious food resource in many parts of the world, is highly valued economically, with its price positively correlated with age. The objective of this study is to construct linear regression models to predict abalone age based on easily measurable physical characteristics.

Data Used:

Reading dataset : Importing the file on dataset in R and reading dataset

```
abalone <- read.csv("abalone.data")
```

Description: df [6 × 9]

	M <chr>	X0.455 <dbl>	X0.365 <dbl>	X0.095 <dbl>	X0.514 <dbl>	X0.2245 <dbl>	X0.101 <dbl>	X0.15 <dbl>	X15 <int>
1	M	0.350	0.265	0.090	0.2255	0.0995	0.0485	0.070	7
2	F	0.530	0.420	0.135	0.6770	0.2565	0.1415	0.210	9
3	M	0.440	0.365	0.125	0.5160	0.2155	0.1140	0.155	10
4	I	0.330	0.255	0.080	0.2050	0.0895	0.0395	0.055	7
5	I	0.425	0.300	0.095	0.3515	0.1410	0.0775	0.120	8
6	F	0.530	0.415	0.150	0.7775	0.2370	0.1415	0.330	20

6 rows

Installing required libraries

```
library(readr)
library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag
```

```
## The following objects are masked from 'package:base':  
##  
## intersect, setdiff, setequal, union  
  
library(tidyr)  
library(ggplot2)  
library(caret)  
  
## Loading required package: lattice  
  
library(reshape2)  
  
##  
## Attaching package: 'reshape2'  
  
## The following object is masked from 'package:tidyr':  
##  
## smiths
```

Explore the structure of the dataset

Structuring the dataset to examine its characteristics, including the variables it contains, their types, the number of observations, and any missing values or outliers.

```
str(abalone)  
  
## 'data.frame': 4176 obs. of 9 variables:  
## $ M : chr "M" "F" "M" "I" ...  
## $ X0.455 : num 0.35 0.53 0.44 0.33 0.425 0.53 0.545 0.475 0.55 0.525 ...  
## $ X0.365 : num 0.265 0.42 0.365 0.255 0.3 0.415 0.425 0.37 0.44 0.38 ...  
## $ X0.095 : num 0.09 0.135 0.125 0.08 0.095 0.15 0.125 0.125 0.15 0.14 ..  
.  
## $ X0.514 : num 0.226 0.677 0.516 0.205 0.351 ...  
## $ X0.2245: num 0.0995 0.2565 0.2155 0.0895 0.141 ...  
## $ X0.101 : num 0.0485 0.1415 0.114 0.0395 0.0775 ...  
## $ X0.15 : num 0.07 0.21 0.155 0.055 0.12 0.33 0.26 0.165 0.32 0.21 ...  
## $ X15 : int 7 9 10 7 8 20 16 9 19 14 ...
```

Data Types:

M: Character (categorical)

X0.455, X0.365, X0.095, X0.514, X0.2245, X0.101, X0.15: Numeric

X15: Integer

Observations:

There are 4176 observations in total.

Data Cleaning and Transformation:

Data cleaning and transformation are crucial steps in the data analysis process, aimed at ensuring that the dataset is accurate, complete, and formatted appropriately for analysis.

Renaming columns

The dataset's columns have been renamed to enhance clarity and understanding. This involved assigning new names to the various variables in the data. By doing so, it simplifies the process for users to comprehend the meaning of each part of the dataset. These alterations improve clarity and ensure that individuals can interpret and analyze the information without confusion.

```
abalone <- abalone %>%
  rename(Sex = M,
         Length = X0.455,
         Diameter = X0.365,
         Height = X0.095,
         Whole_weight = X0.514,
         Shucked_weight = X0.2245,
         Viscera_weight = X0.101,
         Shell_weight = X0.15,
         Rings = X15)
```

Description: df [6 x 9]

	Sex <chr>	Length <dbl>	Diameter <dbl>	Height <dbl>	Whole_weight <dbl>	Shucked_weight <dbl>	Viscera_weight <dbl>	Shell_weight <dbl>	Rings <int>
1	M	0.350	0.265	0.090	0.2255	0.0995	0.0485	0.070	7
2	F	0.530	0.420	0.135	0.6770	0.2565	0.1415	0.210	9
3	M	0.440	0.365	0.125	0.5160	0.2155	0.1140	0.155	10
4	I	0.330	0.255	0.080	0.2050	0.0895	0.0395	0.055	7
5	I	0.425	0.300	0.095	0.3515	0.1410	0.0775	0.120	8
6	F	0.530	0.415	0.150	0.7775	0.2370	0.1415	0.330	20

6 rows

Check for missing values

Checking for missing values in a dataset is essential to facilitate data preprocessing and ensure data quality.

```
summary(is.na(abalone))
```

```
##      Sex      Length      Diameter      Height
## Mode :logical Mode :logical Mode :logical Mode :logical
## FALSE:4176   FALSE:4176   FALSE:4176   FALSE:4176
## Whole_weight Shucked_weight Viscera_weight Shell_weight
## Mode :logical Mode :logical Mode :logical Mode :logical
## FALSE:4176   FALSE:4176   FALSE:4176   FALSE:4176
##      Rings
```

```
## Mode :logical
## FALSE:4176
```

identifying class of data set

```
abalone.class <- sapply(abalone, class)
abalone.class

##           Sex           Length           Diameter           Height  Whole_weight
## "character"    "numeric"      "numeric"         "numeric"    "numeric"
## Shucked_weight Viscera_weight Shell_weight           Rings
## "numeric"      "numeric"      "numeric"         "integer"
```

Explanatory Data Analysis (EDA):

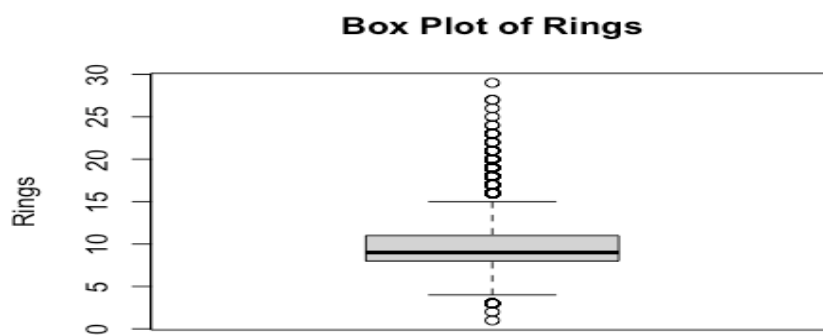
Conducting a thorough EDA to understand the distributions, relationships, and trends within the data. Using visualizations such as scatter plots, histograms, and box plots to examine the relationships between the dependent and independent variables.

Rings Distribution:

“Rings” variable, it serves as the target variable in the dataset. Each integer value in the “Rings” variable corresponds to the number of rings observed on an abalone’s shell. Converting this value to years is achieved by adding 1.5 to the integer value, providing an approximate age in years. Understanding the distribution of the “Rings” variable is crucial for model development and analysis. It helps in identifying the range and spread of ring counts, which are essential factors for predicting abalone age accurately.

Box Plot of Rings

```
boxplot(abalone$Rings, main = "Box Plot of Rings", ylab = "Rings")
```



Description: The box plot provides a visual summary of the central tendency, spread, and potential outliers in the dataset, offering insights into the distribution of abalone rings where the mean ranges between 5-10 rings, with the minimum and maximum number of rings set at 1 and 29, respectively.

Creating intervals

: Creating intervals for the number of rings in abalone to visualize the distribution of ring counts across the dataset.

```
intervals <- c(0, 5, 10, 15, 20, 25, 30)
intervals

## [1]  0  5 10 15 20 25 30

interval_labels <- paste(intervals[-length(intervals)], intervals[-1], sep =
"-")
interval_labels

## [1] "0-5"  "5-10" "10-15" "15-20" "20-25" "25-30"
```

Ring Frequency table:

The output displays a frequency table of the intervals created for the number of rings in abalone. Each interval represents a range of ring counts, and the corresponding frequencies indicate how many abalones fall within each interval.

```
ring_intervals <- cut(abalone$Rings, breaks = intervals, right = FALSE)
frequency_table <- table(ring_intervals)
print(frequency_table)

## ring_intervals
##  [0,5) [5,10) [10,15) [15,20) [20,25) [25,30)
##      74   2022   1717    301     57      5
```

Convert interval_labels to factor with desired order of levels

```
ring_frequencies <- hist(abalone$Rings, breaks = intervals, plot = FALSE)$counts
```

Pie chart of rings distribution

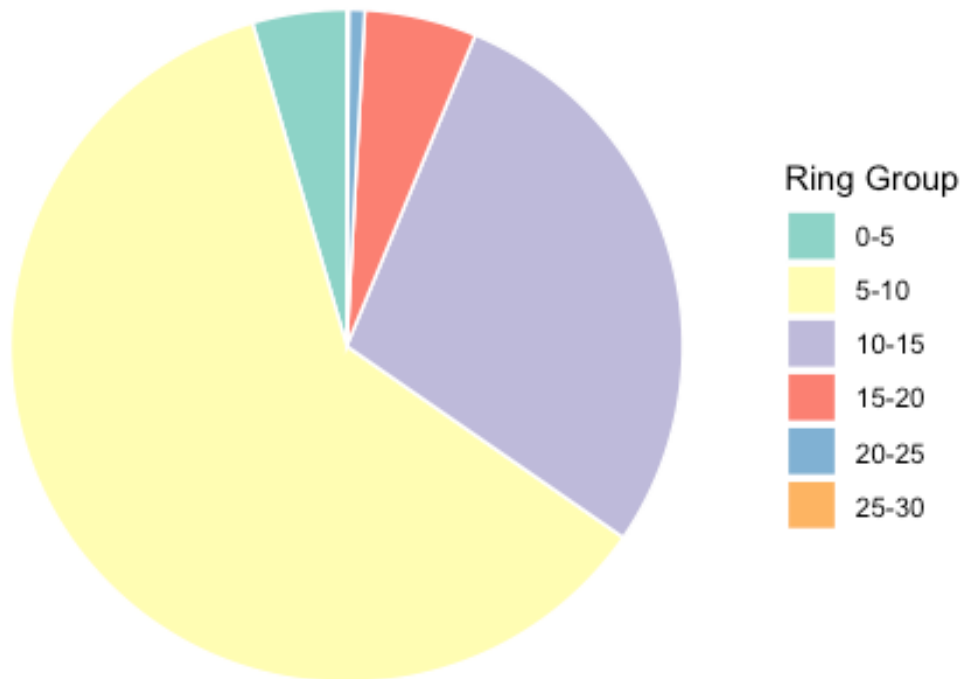
```
pie_data <- data.frame(interval = interval_labels, frequency = ring_frequencies)
pie_data$interval <- factor(pie_data$interval, levels = interval_labels)
pie_data$interval

## [1] 0-5  5-10 10-15 15-20 20-25 25-30
## Levels: 0-5 5-10 10-15 15-20 20-25 25-30

pie_chart <- ggplot(pie_data, aes(x = "", y = frequency, fill = interval)) +
  geom_bar(stat = "identity", width = 1, color = "white") +
  coord_polar("y", start = 0) +
  scale_fill_brewer(palette = "Set3") +
  theme_void() +
  theme(legend.position = "right") +
```

```
labs(title = "Distribution of Rings in Abalone", fill = "Ring Group")
print(pie_chart)
```

Distribution of Rings in Abalone



Description: The pie chart illustrates the distribution of ring counts among abalones, categorized into intervals. Each interval represents a range of ring counts, and the corresponding frequencies indicate the number of abalones falling within each interval. Through the pie chart, one can visually assess the proportion of abalones falling within each ring count interval, providing a clear depiction of the distribution pattern within the dataset.

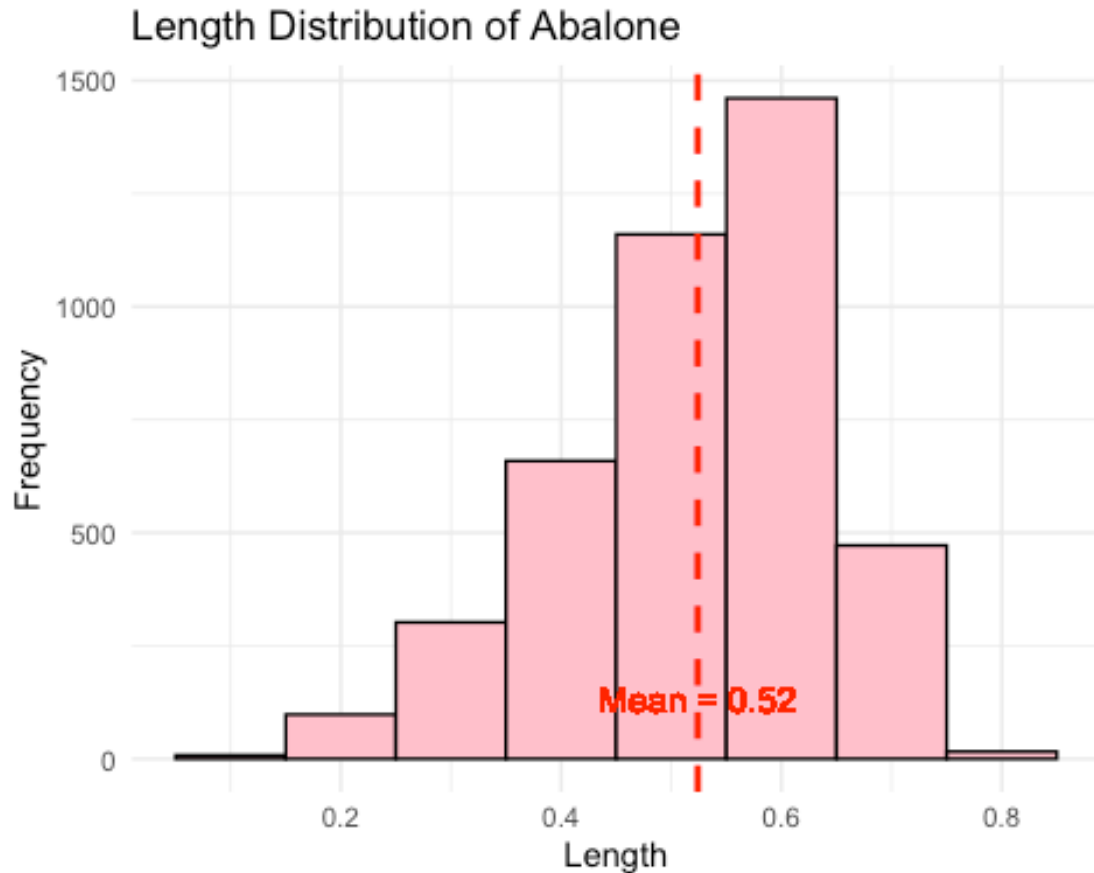
Mean Distribution lenght

```
mean_Length <- mean(abalone$Length, na.rm = TRUE)
print(mean_Length)

## [1] 0.5240086

ggplot(abalone, aes(x = Length)) +
  geom_histogram(binwidth = 0.1, fill = "pink", color = "black") +
  geom_vline(xintercept = mean_Length, color = "red", linetype = "dashed", size = 1) +
  geom_text(aes(x = mean_Length, label = paste("Mean =", round(mean_Length, 2))),
    y = 50, vjust = -1, color = "red", size = 4) +
```

```
labs(title = "Length Distribution of Abalone",
     x = "Length",
     y = "Frequency") +
theme_minimal()
```



Description: By visualizing the mean distribution, we can assess the overall variability and dispersion of abalone lengths within each subset, aiding in further analysis and interpretation.

Training the dataset to evaluate the performance of modeling

```
trainingSet = read.csv('abalone.data')
head(trainingSet)
```

```
##  M X0.455 X0.365 X0.095 X0.514 X0.2245 X0.101 X0.15 X15
## 1  M  0.350  0.265  0.090  0.2255  0.0995  0.0485  0.070  7
## 2  F  0.530  0.420  0.135  0.6770  0.2565  0.1415  0.210  9
## 3  M  0.440  0.365  0.125  0.5160  0.2155  0.1140  0.155  10
## 4  I  0.330  0.255  0.080  0.2050  0.0895  0.0395  0.055  7
## 5  I  0.425  0.300  0.095  0.3515  0.1410  0.0775  0.120  8
## 6  F  0.530  0.415  0.150  0.7775  0.2370  0.1415  0.330  20
```


Building linear regression model in between rings and length of abalone correlation coefficient

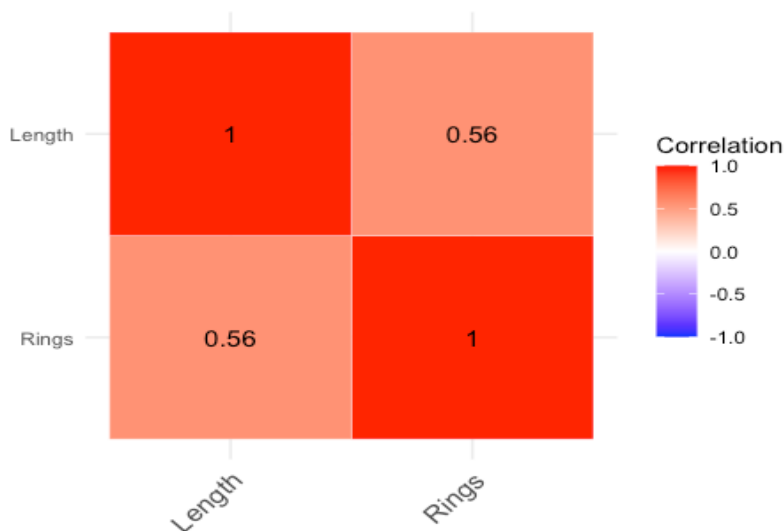
```
correlation_coefficient <- cor(abalone$Rings, abalone$Length)
correlation_coefficient

## [1] 0.5571226
```

The **correlation coefficient** between “Rings” and “Length” in the abalone dataset is approximately 0.557. This positive correlation coefficient suggests that there is a moderate positive linear relationship between the number of rings and the length of abalones. Fit linear regression model

Creating correlation matrix

```
correlation_matrix <- cor(abalone[c("Rings", "Length")])
correlation_df <- as.data.frame(correlation_matrix)
correlation_df$Var1 <- rownames(correlation_df)
melted_df <- melt(correlation_df, id.vars = "Var1", variable.name = "Var2", value.name = "Correlation")
ggplot(data = melted_df, aes(x = Var1, y = Var2, fill = Correlation)) +
  geom_tile(color = "white") +
  scale_fill_gradient2(low = "blue", mid = "white", high = "red", midpoint = 0, limit = c(-1, 1), space = "Lab", name = "Correlation") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, vjust = 1, size = 12, hjust = 1)) +
  labs(x = "", y = "") + # Remove axis labels
  geom_text(aes(label = round(Correlation, 2)), color = "black", size = 4) +
  # Add labels
  coord_fixed()
```



Description: The correlation matrix between the “Rings” and “Length” of abalone represents the pairwise correlations between these two variables. Each cell in the matrix contains a correlation coefficient that quantifies the strength and direction of the linear relationship between “Rings” and “Length”. the s positive (closer to +1), indicates a positive linear relationship, meaning that as one variable (“Length”) increases, the other variable (“Rings”) tends to increase as well.

Modeling:

Splitting the dataset into training and testing sets to evaluate the performance of the model. Developing a linear regression model using the selected features. Also, considering building a models to compare two sets of features.

```
model <- lm(Rings ~ Length, data = abalone)
model

##
## Call:
## lm(formula = Rings ~ Length, data = abalone)
##
## Coefficients:
## (Intercept)      Length
##      2.097      14.953

summary(model)

##
## Call:
## lm(formula = Rings ~ Length, data = abalone)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.9660 -1.6973 -0.7417  0.8751 16.6789
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   2.0968     0.1855   11.30  <2e-16 ***
## Length       14.9534     0.3450   43.34  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.677 on 4174 degrees of freedom
## Multiple R-squared:  0.3104, Adjusted R-squared:  0.3102
## F-statistic: 1879 on 1 and 4174 DF, p-value: < 2.2e-16
```

MODEL SUMMARY

Call: This section shows the call that was used to create the linear regression model. It specifies the formula used (Rings ~ Length) and the dataset (abalone) that was used for fitting the model.

Coefficients: The coefficients section provides the estimated coefficients for the intercept and the predictor variable (Length).

Intercept: The estimated intercept coefficient is 2.097. This is the predicted number of rings when the length of the abalone is zero. However, this value may not have a practical interpretation since abalone cannot have a length of zero.

Length: The estimated coefficient for the Length predictor variable is 14.953. This indicates that for each unit increase in the length of the abalone, the predicted number of rings is expected to increase by approximately 14.953 units.

RESULT OF FITTED MODEL

Intercept: The intercept term is estimated to be 2.097. This represents the predicted value of the response variable ("Rings") when the predictor variable ("Length") is zero. However, in practical terms, this interpretation might not make sense depending on the context of the data.

Length: The coefficient for the predictor variable "Length" is estimated to be 14.953. This suggests that for each unit increase in the length of the abalone, the predicted number of rings increases by approximately 14.953 units.

Significance: Both the intercept and the coefficient for "Length" are highly significant, as indicated by the very small p-values (typically < 0.05), denoted by the '***' symbols.

R-squared: The R-squared value of 0.3104 indicates that approximately 31.04% of the variance in the number of rings can be explained by the linear relationship with the length of the abalone.

Adjusted R-squared: The adjusted R-squared value of 0.3102 adjusts for the number of predictors in the model.

F-statistic: The F-statistic of 1879 with an associated very small p-value indicates that the model as a whole is highly significant.

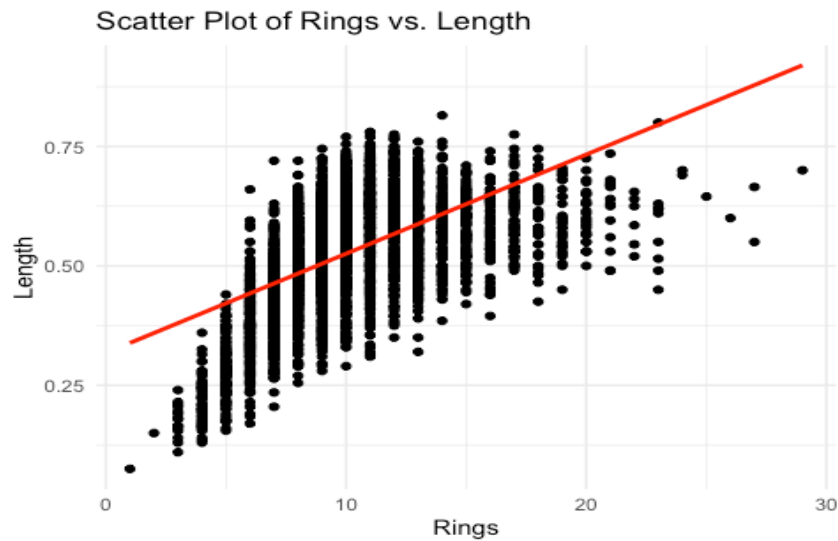
In summary, the linear regression model suggests that the length of the abalone is a significant predictor of the number of rings, and approximately 31% of the variance in the number of rings can be explained by the length of the abalone alone.

Linear regression plot between rings and length

```
ggplot(abalone, aes(x = Rings, y = Length)) +  
  geom_point() + # Add points  
  geom_smooth(method = "lm", se = FALSE, color = "red") + # Add Linear regression line  
  labs(x = "Rings", y = "Length") + # Label axes
```

```
ggtitle("Scatter Plot of Rings vs. Length") + # Add title
theme_minimal()

## `geom_smooth()` using formula = 'y ~ x'
```



Description: scatter plot shows a linear regression line overlaid, visualizing the relationship between the number of rings and the length of abalone specimens. This plot helps in understanding how the length of an abalone relates to the number of rings it possesses, potentially indicating growth patterns or other biological relationships.

Evaluation and Results:

Evaluating the performance of the model using metrics, such as R-squared, Mean Absolute Error (MAE), and Mean Squared Error (MSE).

```
predictions <- predict(model)

# Calculate residuals
residuals <- abalone$Rings - predictions

# Calculate R-squared
rsquared <- summary(model)$r.squared
print(paste("R-squared:", rsquared))

## [1] "R-squared: 0.310385572625062"

# Calculate Mean Absolute Error (MAE)
mae <- mean(abs(residuals))
print(paste("Mean Absolute Error (MAE):", mae))

## [1] "Mean Absolute Error (MAE): 1.95049213480304"
```

```
# Calculate Mean Squared Error (MSE)
mse <- mean(residuals^2)
print(paste("Mean Squared Error (MSE):", mse))

## [1] "Mean Squared Error (MSE): 7.16448568172938"
```

Result driven:

R-squared (R^2): R-squared measures the proportion of the variance in the dependent variable (Rings) that is predictable from the independent variable (Length) in the model. In this case, R-squared is approximately 0.31, which means that around 31% of the variance in Rings can be explained by the linear relationship with Length. Mean Absolute Error (MAE): MAE measures the average absolute difference between the actual values and the predicted values. Here, MAE is approximately 1.95. It means, on average, the model's predictions are off by around 1.95 units of Rings. Mean Squared Error (MSE): MSE measures the average squared difference between the actual and predicted values. In this case, MSE is approximately 7.16. It's larger than MAE because it penalizes larger errors more heavily due to squaring. Interpretation:

Conclusion:

In conclusion, the analysis reveals valuable insights into the relationship between the number of rings and the length of abalone, key factors in determining the age of this resourceful marine species.

The correlation coefficient between the number of rings and the length of abalone is approximately 0.557, indicating a moderate positive linear relationship. This suggests that as the length of abalones tends to increase, the number of rings also tends to increase, though not extremely strongly.

Moreover, the linear regression model confirms this relationship, with each unit increase in the length of an abalone corresponding to a predicted increase of approximately 14.9534 units in the number of rings.

Overall, these findings offer valuable contributions to understanding and predicting the age of abalone, providing insights that can aid in their management and utilization as a valuable food resource.